

Drift Happens

Intelligent Ways to Detect and Remediate It



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Who's this guy?



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What is Drift?

- Design vs IRL
- Git vs IRL
- Manual changes
- Emergency fixes
- Failed deployments



Who Cares?

- EVERYONE
- Compliance
- Finance
- Security
- Development
- Support Teams

Drift Types

Not all drift is bad, but all drift needs explaining.

CONFIGURATION

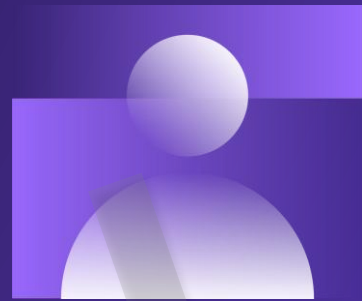
GOVERNANCE

OPERATIONAL

ACCEPTED

What Causes Drift In Azure?

Most drift is not malicious



People

- “just this once”
- Outages
- Urgent support
- Unclear IaC ownership
- Lack of IaC skill
- Separate teams
- Shared platforms



Policy

- New policies
- Updated policies
- Changes to baseline



Pipelines

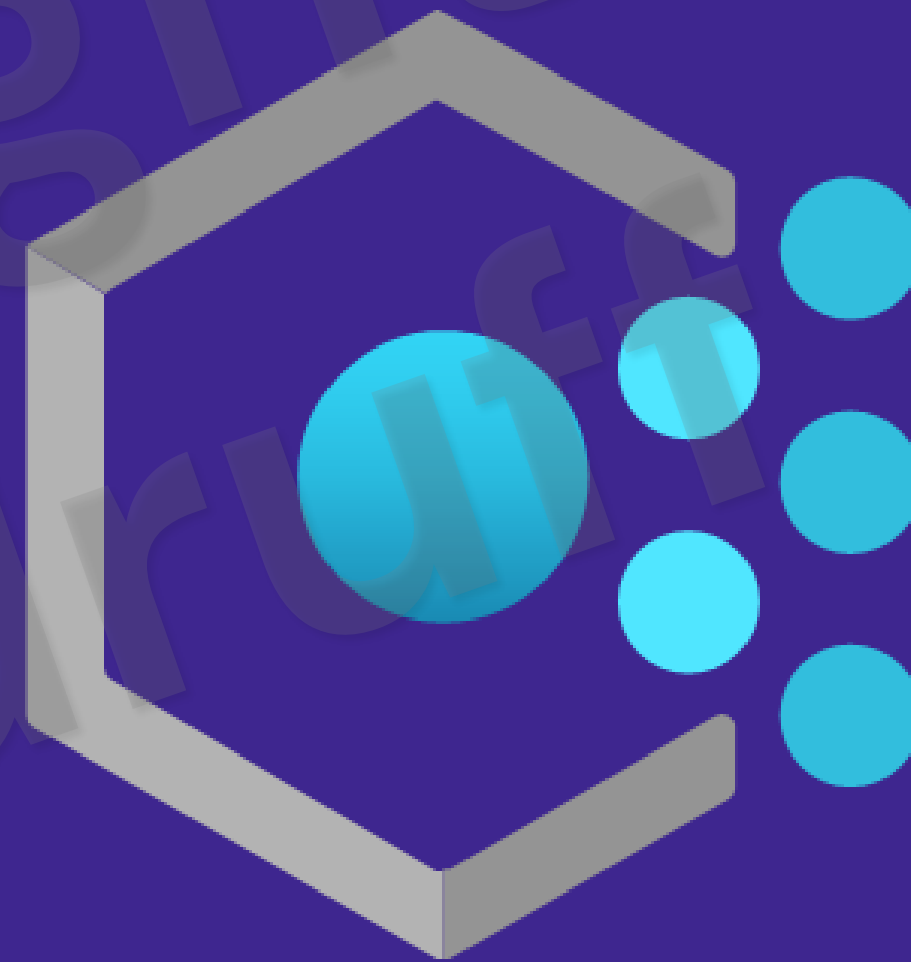
- Failed or partial runs
- Application delivery
- Skipped environments
- Changes outside the PR flow

How Can We Control Drift?

Daily checks won't cut it, automation is required!

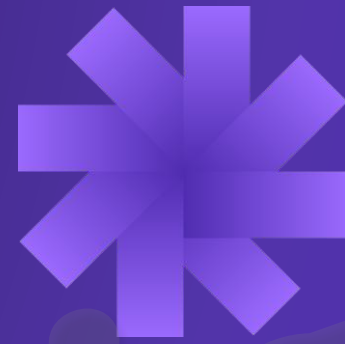
Azure Policy As A Guardrail

- Define what is allowed, what must exist and what should be flagged
- Audit, deny, modify or deployIfNotExists
- Prevent daft changes with `deny`
- Remediate resources with automation accounts
- Policy will not replace IaC, but it is an excellent guardrail for things you never want to rely on humans remembering.



What is Infrastructure as Code?

The benefits are huge, if it makes sense for your business



Git Integrated

Code is stored for repeatable deployment, if the code changes so does the infra



Deployment method

IaC is a way to provision and manage infra at scale using code



Resource Declaration

Code describes infrastructure declaratively (e.g., Terraform) or imperatively (e.g., scripts)

Common IaC Tools

Terraform –
Multi-platform, HCL

ARM/Bicep –
Azure-native,
JSON/YAML-based

Pulumi –
IaC in real
programming
languages

PowerShell –
Native Microsoft,
easiest option

CloudShell –
Native Microsoft,
latest script tech

Ansible/Puppet/Chef
– Configuration
management

Automation Without Regret

- Code-defined infrastructure
- Trusted state management
- Automated detection and alerting
- Intelligent summarisation and routing
- Human-gated response

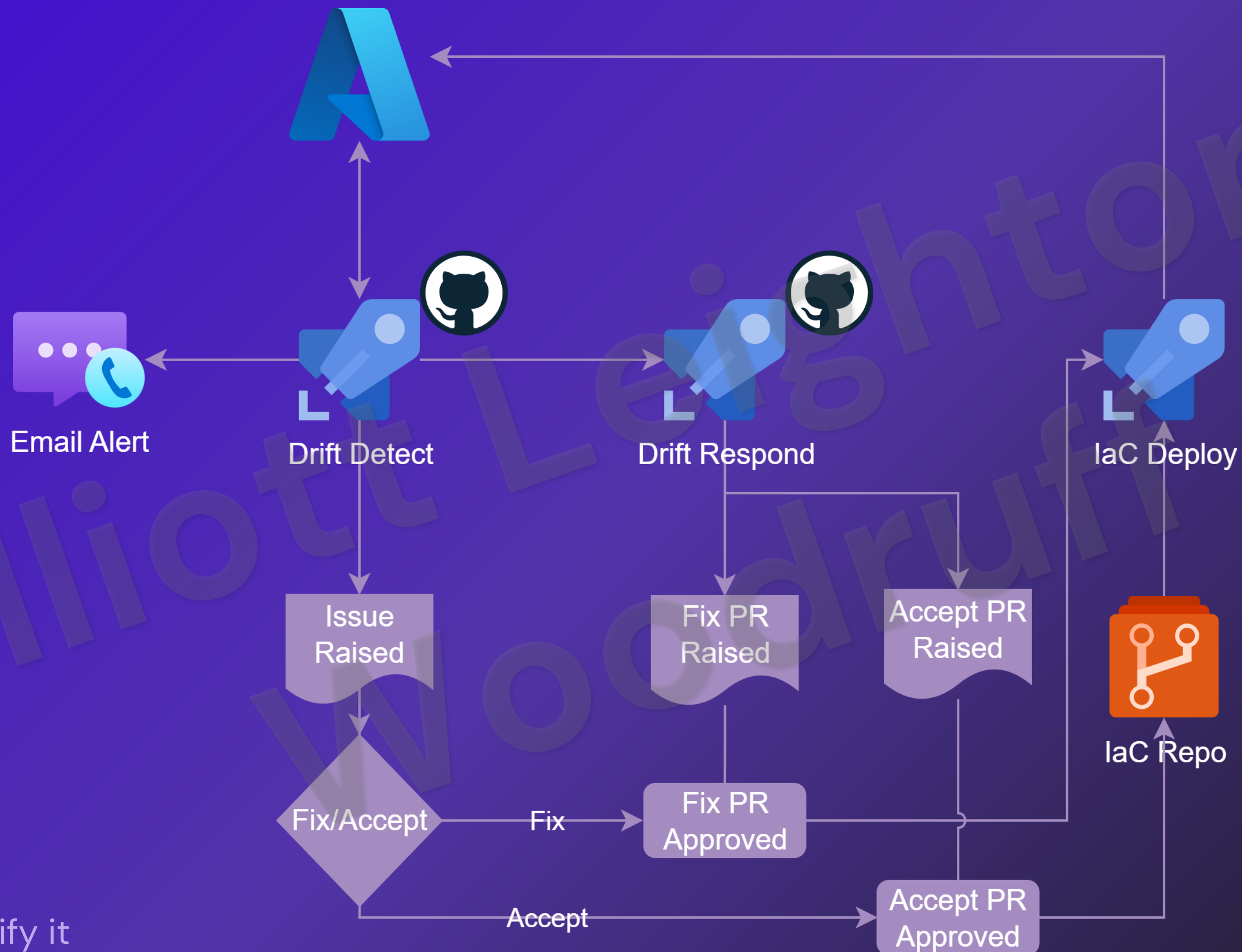


IaC Deploy



IaC Repo

Detect it
Classify it
Route it
Fix it or codify it



Detective Inspector Drift

Thirty-two years on the beat
Grumpy but highly skilled





Agent Remedy

Remediation Division
Dispatched by the Inspector

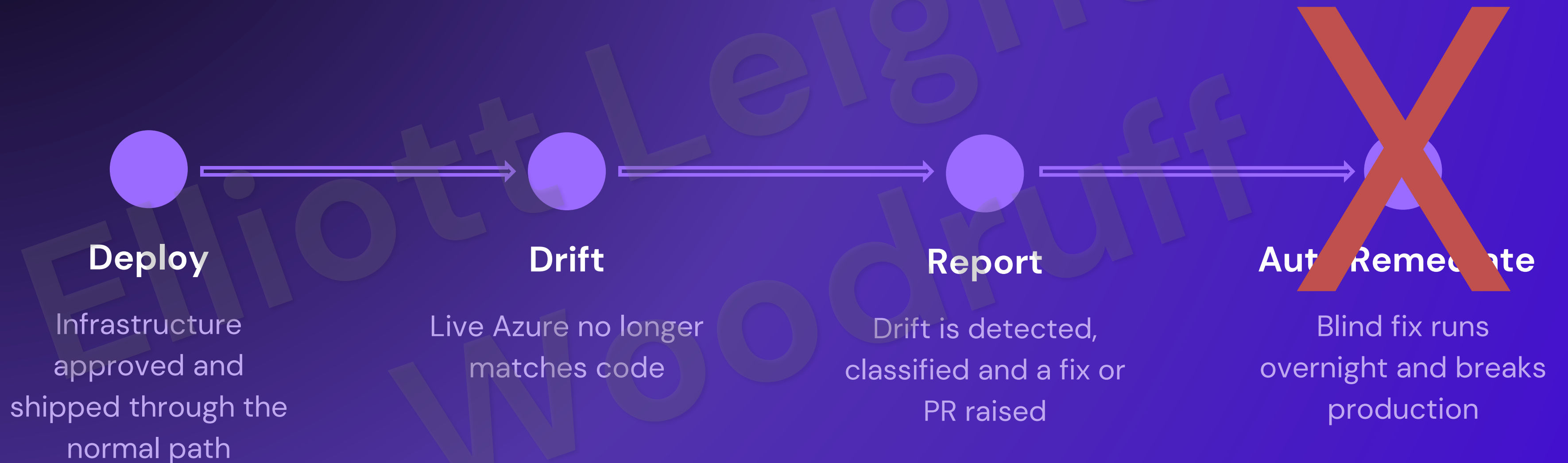


Demo Time



Risky Automation

Know where to stop....



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Thank you!

Q&A Time

